

Simulate message flow between Media Service and Audio HAL using Android logs or a flowchart. Demonstrate how user input propagates through the stack

Module 6

IVI In-Vehicle Infotainment Systems

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PROJECT TYPE: Conceptual Simulation / System Study

ABSTRACT

In-Vehicle Infotainment (IVI) systems provide important functions such as- media playback, navigation, communication, and connectivity inside modern vehicles. This project focuses on the simplified message flow between the Media Service and Audio Hardware Abstraction Layer (Audio HAL) in an IVI system.

The project demonstrates how a user's media command, such as Play, Pause, or Resume, can propagate through different software layers before reaching the audio hardware. Android-style logs are used to simulate this communication and observe the sequence of events.

The simulation covers three basic operations: Play, Pause, and Resume. The results show the movement of user commands from the Media Interface to the Media Service, Audio Framework, Audio HAL, and finally the audio hardware. This project is a conceptual simulation and does not involve real vehicle hardware, vehicle buses, or an actual Android Automotive deployment.

INTRODUCTION

- i. **What an IVI system is and why it matters?**

- ➔ An In-Vehicle Infotainment (IVI) system combines entertainment, information, communication, and connectivity features inside a vehicle. It supports functions such as music playback, navigation, phone connectivity, and user interaction. A well-integrated IVI system improves the driver experience and helps provide information and controls in a convenient manner.
- ii. **Objective of the chosen option**
 - ➔ The objective of this project is to simulate the message flow between the Media Service and Audio HAL. The project demonstrates how user commands such as Play, Pause, and Resume are passed through the Audio Framework before reaching the audio hardware.
- iii. **Scope and Limitations**
 - ➔ The project focuses on a simplified simulation of the IVI audio message flow using Android-style logs. It is intended for understanding the interaction between software layers. No real vehicle hardware, vehicle communication bus, or physical Android Automotive system is used.

BACKGROUND / LITERATURE

IVI system uses different software layers to manage media and audio operations. The Media Service handles user media request, while the Audio Framework manages audio operations. The Audio HAL provides an interface between the software and audio hardware. These layers work together to process commands such as Play, Pause, and Resume.

TOOLS AND ENVIRONMENT

- Microsoft Word – Report Preparation
- Text Editor – Creating simulated log data
- GitHub – Project storage and submission
- Simulated Android-style logs – Demonstrating message flow

SYSTEM DESIGN / METHODOLOGY

The project follows a simplified message flow in which a user command is passed through different IVI software layers.

User Input -> Media Service -> Audio Framework -> Audio HAL -> Audio Hardware

For this simulation, three commands are considered: Play, Pause, and Resume. The message flow for each command is represented using simulated Android-style logs.

IMPLEMENTATION / MODELLING

The message flow was modelled by using simulated Android-style logs. The logs represent how user commands move through the IVI audio stack.

PLAY COMMAND

User Input: PLAY

Media UI -> Media Service: PLAY_REQUEST

Media Service -> Audio Framework: START_AUDIO

Audio Framework -> Audio HAL: OPEN_OUTPUT

Audio HAL -> Audio Hardware: AUDIO_PLAYING

PAUSE COMMAND

User Input: PAUSE

Media UI -> Media Service: PAUSE_REQUEST

Media Service -> Audio Framework: PAUSE_AUDIO

Audio Framework -> Audio HAL: PAUSE_OUTPUT

Audio HAL -> Audio Hardware: AUDIO_PAUSED

RESUME COMMAND

User Input: RESUME

Media UI -> Media Service: RESUME_REQUEST

Media Service -> Audio Framework: RESUME_AUDIO

Audio Framework -> Audio HAL: RESUME_OUTPUT

Audio HAL -> Audio Hardware: AUDIO_PLAYING

These logs demonstrate the propagation of user input through the simplified IVI audio stack.

TESTING AND RESULTS

The simulated message flow was tested using three basic media commands:

SERIAL NUMBER	TEST CASE	USER INPUT	EXPECTED RESULT	STATUS
1.	T1	Play	Audio starts	PASS
2.	T2	Pause	Audio pauses	PASS
3.	T3	Resume	Audio resume	PASS

The simulated logs showed the expected movement of each command through the Media Service, Audio Framework, and Audio HAL.

DISCUSSION

The simulation demonstrates how a user's media command is processed through different IVI audio layers. The simulated logs show that Play, Pause, and Resume commands reach the Audio HAL in the expected sequence. This Simplified model helps in understanding the interaction between software services and audio hardware.

CONCLUSION AND FUTURE SCOPE

The project successfully demonstrated the simplified message flow between the Media Service and Audio HAL using simulated Android-style logs. Play, Pause, and Resume commands were successfully represented through the IVI audio stack.

In the future, the model can be tested using an Android Automotive Emulator or real IVI hardware for practical validation.

REFERENCES

1. Android Open-Source Project, "Android Automotive Audio Overview."
2. Android Open-Source Project, "Audio Control HAL."
3. Android Developers, "Extend your media app to Android for Cars."

APPENDIX

Simulated Android-Style Logs

PLAY -> Media Service -> Audio Framework -> Audio HAL -> Audio Hardware

PAUSE -> Media Service -> Audio Framework -> Audio HAL -> Audio Hardware

RESUME -> Media Service -> Audio Framework -> Audio HAL -> Audio Hardware

These logs are simulated for academic demonstration and were not captured from a real Android Automotive system.